

MICROGRID

Modular Integrated Campus Resource Optimisation and Geospatial Representation for Intelligent Distribution

Demonstration Area	Demo	Business Use Case
Arnhem's Buiten	Dutch pilot	TOPIC 8

Participant

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Abstract

In the Netherlands and the EU, (business) campuses face a growing set of challenges related to energy management. Due to increasing grid congestion, expanding or upgrading connections is often difficult or even impossible, forcing campuses to make efficient use of their existing infrastructure. Simultaneously, they must ensure regulatory compliance while pursuing cost savings, energy efficiency, and sustainability goals as part of the broader energy transition. These campuses also need to manage a diverse mix of energy carriers, including electricity, natural gas and heat, while maintaining reliability and transparency across all stakeholders on the campus.

To address these challenges, the campus (energy) managers need

- A shared, decision-ready view of the campus energy system
- To simulate and predict what-if scenario's.

We propose a digital-twin solution called MICROGRID for visualising energy consumption, distribution, division and simulation on (business) campuses using the HedgeIoT framework. We will provide an interactive and intuitive way to explore and understand the energy system of the Arnhem's Buiten Electricity Campus. The solution has three main components:

1. **A 3D Digital Twin Interface.** A web-based three-dimensional model of the campus showing both the spatial distribution and functional division of energy flows (electricity, gas, heat, etc.). Users can navigate between buildings and systems to see how energy moves across the site. We were inspired by the [3DBAG-initiative](#).
2. **Data Connection with HedgeIoT.** The application will connect to live data streams from the HedgeIoT framework (through ETSI SAREF / Knowledge Engine (KE) interfaces). These data streams will feed real-time information into the 3D model, enabling dynamic updates of usage patterns, alerts, and system status for Distributed Energy Resources (DERs).
3. **Simulation and Alert Visualisation.** Basic simulation capabilities will allow users to explore "what-if" scenarios, such as adjusting demand or adding new DERs to understand possible outcomes. In addition, the system will visualise anomalies, alerts, and other insights coming from the HedgeIoT network, helping users identify inefficiencies or issues in an intuitive geospatial way.



The demonstrator will show how digital-twin visualisation using HedgeIoT's semantic interoperability can improve understanding, transparency and decision-making around local energy systems. Within the project, we will deliver a working demonstrator integrated with HedgeIoT data streams. By combining intuitive 3D visualisation, real-time data and simulation, this project contributes a replicable and user-friendly interface layer for campus-level energy management, helping accelerate the digitalisation and optimisation of distributed energy systems.

Free Keywords: Digital Twin, Energy Visualisation, 3D Web Application, HedgeIoT, Semantic Interoperability, Campus Energy Systems, What-If Simulation, Alert Visualisation

We propose to use this open call as a stepping stone for wider deployment across other Closed Distribution System (CDS) campuses (Dutch: [Gesloten Distributiesysteem](#)), as defined by the Dutch Authority for Consumers and Markets (ACM). They many face similar problems as Arnhem's Buiten. These (~150) campuses have a legal obligation to [register](#) and are thus easily contacted.

1. Technical Quality

Web tech stack: We will develop the MICROGRID demonstrator using a modern and widely adopted technology stack consisting of React, Node.js, TypeScript, and PostgreSQL. At Bronscore, we focus on this web-technology stack, establishing ourselves as experts with a proven track record of delivering production-grade web applications.

Web-application: Bronscore specialises in building real-world complex web-applications. We make use of our framework [Eidos](#) to develop this application. This framework uses type-theory as the basis for quickly generating parts of the user interface needed for common functional management activities.

3D visualisation: We shall leverage the results of the "[3DBAG](#)" dataset developed by the Technical University of Delft to create the 3D-visualization. We have inspected this dataset and concluded that it will be suitable for our use case. What remains is mapping the ontology onto accurate spatial coordinates, and creating an intuitive editing interface for the ontology in this new paradigm.

HedgeIoT connectors: We have foundational understanding of the current software architecture at the campus and the principles behind HedgeIoT and the knowledge engine. We shall work closely with the current technology partners at Arnhem's Buiten, such as TNO, to create a well-adapted demonstrator. Even though there is a lot of information available online on ETSI SAREF and the knowledge engine, many aspects regarding the adapters available at the campus and the availability of datasets are still unknown to us (see technological risks). Given our academic background and experience with different data-formats we believe this will not be a hurdle but rather an opportunity to test whether experienced programmers without prior familiarity with SAREF can effectively work with it.

Real-time visualisation: We will use socket.io websockets to communicate between the backend and frontend. We have worked with this technology extensively in the past. We have experience with creating complex [visualisations](#). We will use graphing libraries such as Visx to speed up development.

Portability: Given our goal of productising the MICROGRID application, it is important to build it in such a way as to enable adaptability for multiple business campuses. We currently maintain software applications for multiple customers leveraging techniques like feature flags, extensive git-branching, CI/CD pipelines and configurability.

Simulation: We will aim for simulation of basic fictive DERs in what-if analyses using publicly available data and basic data augmentation methods. The problem of creating realistic synthetic datasets is outside of the scope for this project, but in the future such datasets can easily be injected into our solution.

1.1. Alignment

MICROGRID directly addresses Topic 8 by implementing value-adding services on top of the HEDGE-IoT Semantic Interoperability Framework (SIF) and Knowledge Engine for the Arnhem's Buiten Electricity Campus BUC. Concretely, we (i) deliver role-based, real-time dashboards for distinct stakeholder groups (facility operator, tenant, grid/DSO liaison), and (ii) provide a campus digital twin that integrates live data via SAREF-compliant semantic adapters.

Where anomaly-detection services are available within HEDGE-IoT, MICROGRID consumes and visualises them; where not, it exposes the hooks and UI patterns for seamless future inclusion. All integrations follow HEDGE-IoT architectural guidelines (semantic mapping, adapter contracts, authn/authz), enabling reuse across other Closed Distribution Systems (CDS). Alignment KPIs: ≥ 5 distinct asset types mapped to the campus ontology; < 60 s end-to-end telemetry-to-UI latency; ≥ 2 role-specific dashboards; SAREF mapping validated with campus partner; demonstrator deployed on-site and exercised in an operational walkthrough.

1.2. Objectives

Overall objective: deploy a SAREF-aligned, web-based digital twin for Arnhem's Buiten that turns heterogeneous campus energy data into actionable, role-specific insights for operations and planning.

1. Semantic integration & data flow. Connect ≥ 3 live data streams (e.g., substation/metering, building loads, DER) through SIF/Knowledge Engine; deliver translation layers and mapping docs; target <60s ingest-to-visualisation latency. Acceptance: integration test report; mapping catalogue; latency measurements.
2. Role-based visualisation. Ship dashboards for facility operator and tenants. Acceptance: UX walkthroughs signed off by campus stakeholders; KPI catalogue.
3. Digital twin & "what-if". Provide a 3D campus view bound to the semantic model, plus lightweight scenario probes (e.g., load shift, DER addition). Acceptance: scenario demo scripts; comparison views; correctness spot-checks with partner data.
4. Alerts & anomaly visualisation. Integrate available HEDGE-IoT anomaly/alert outputs; standardise how alerts are contextualised in space/time and triaged per role. Acceptance: alert taxonomy, sample alert flows, operator-acknowledgement trail.

1.3. Concept and approach / Project Description

MICROGRID will simulate fast "what-ifs" by transforming live + historical time-series (via SIF/KE) instead of running heavy optimisers. The engine decomposes signals with filtering (baseload/cyclic/events), applies windowing, and models actions through time-shifting, reflection, and controlled shuffling to preserve seasonality while testing load shifting. We assume basic dataset availability as mentioned in the BUC. We want to demonstrate (third party) model/alert in-cooperation in the 3D visualisation.

1.4. Ambition

MICROGRID moves HEDGE-IoT from framework to field by uniting three capabilities: semantic interoperability (SAREF-based), spatially explicit 3D visualisation, and interactive scenario exploration. Beyond conventional SCADA dashboards, it gives decision-makers a shared, geo-spatially anchored view of constraints and options, accelerating consensus on flexibility, efficiency, and investment measures. Technically, we aim for TRL 5/6 in a relevant environment, demonstrating low-friction SIF/KE integration patterns, reusable semantic mappings, and a UI that remains robust as ontologies evolve. Strategically, the ambition is a replicable blueprint for the ~150 Dutch CDS campuses (and analogous EU sites), with open documentation and selectively open-sourced components to encourage take-up, while retaining a path to a sustainable SaaS product. The result is a credible, standards-based reference demonstrator that strengthens the market case for interoperable, data-driven campus energy management.

2. Impact

2.1. Expected impact and results

The MICROGRID demonstrator will strengthen the business case for interoperable, data-driven energy management on campuses and comparable local energy systems. By combining semantic interoperability, simulation, and intuitive 3D visualisation, it enables decision-makers to understand both the technical and spatial implications of their energy strategies, rather than relying solely on financial or aggregated data. This combination of economic and geospatial insight helps campus operators identify feasible measures for efficiency, flexibility, and sustainability, while ensuring regulatory compliance and operational reliability.

Beyond its technical and organisational value, the project also generates strong visibility impact. The Arnhem's Buiten Electricity Campus acts as a [national showcase](#) for innovation in the energy

transition, regularly hosting policymakers, utilities, and technology developers. By presenting a visually engaging, real-time digital twin of the local grid, MICROGRID provides a clear and compelling way to communicate how digitalisation and semantic interoperability can make local energy systems more transparent and efficient. Together with campus partners, the demonstrator will be featured at regional and national energy events, reinforcing the position of both Arnhem Buiten and the HEDGE-IoT framework as leading examples of practical, interoperable innovation.

In the longer term, the expected impact includes improved collaboration between operators and technology providers, greater public and stakeholder understanding of energy flows and constraints, and a strengthened market position for interoperable, standards-based visualisation tools. Impact will be measured through the demonstrator's use in operational decision-making, its presence at major industry showcases, interest from other campuses, and integration of its components into ongoing commercial and research initiatives.

2.2. Exploitation plan of project results

The exploitation of the MICROGRID results will follow a three-step roadmap, ensuring continuity beyond the open call and maximising both visibility and replication potential within the HEDGE-IoT ecosystem.

Step 1. Demonstrator deployment and visibility at Arnhem Buiten.

In the first stage, we focus on achieving a fully functional and visually robust demonstrator at the Arnhem Buiten Electricity Campus, which serves as the primary validation environment and a living lab for applied research. This step is not only about technical validation but also about knowledge transfer between research and industry. The MICROGRID team combines strong expertise in data modelling and visualisation with an academic background, ensuring that research concepts around semantic interoperability and digital-twin representation are translated into a working, real-world application.

Unlike a typical business campus, Arnhem Buiten plays a pioneering role in the Dutch energy transition. There are many converging initiatives around this campus, e.g.:

- The Alliander Flexlab (DSO) for Energy Management Systems
- Tennet (TSO), Alliander (DSO), DNV (consultancy), Hyosung
- The Perspective Fund Gelderland (Perspectieffonds Gelderland), €200M
- OostNL, the regional innovation fund, part owner of the campus

This makes the campus the hosting location for many energy related networking events. We aim to showcase the demonstrator at multiple such energy-related events. Our goal is for the MICROGRID visualisation to become a flagship exhibit that demonstrates the value of digital twins for campus energy management, both for experts and the general public.

Through this ecosystem, the MICROGRID demonstrator will act as both a research translation platform and a public showcase. The project team will work closely with campus partners and local knowledge institutions like TNO to ensure scientific insights are applied in an operational context.

Step 2. Replication with a second CDS pilot.

In collaboration with OostNL and the Arnhem's Buiten stakeholders, aim to leverage our visibility to position the demonstrator as a pioneering reference case for other campuses with a closed distribution system.

This second pilot will test interoperability with additional HEDGE-IoT components which will likely need to be installed at the second pilot location. It will represent our first commercial deployment, validating the business potential of the MICROGRID solution and strengthening the bridge between the open-call results and long-term market exploitation.

Step 3. Expansion to other CDS

Building on the Dutch pilot results, the final stage targets scaling across the network of approximately 150 CDS campuses in the Netherlands. Each site can leverage the same semantic integration and 3D visualisation stack, adapted to its own energy topology.

Through collaboration with energy clusters and standardisation bodies, the MICROGRID architecture will be positioned as a reusable blueprint for campus-level digital-twin visualisation within the HEDGE-IoT Semantic Interoperability Framework. The long-term vision is to provide a SaaS solution for visualising and interacting with campuses that are empowered by HedgeIoT.

3. Quality of Work plan

MICROGRID will deliver a practical, working digital twin for the Arnhem's Buiten Electricity Campus in six months, organised around three clear tasks: (T1) connect to SIF, (T2) build an intuitive web app with role-based dashboards and a 3D campus view, and (T3) add simple "what-if" scenarios and alert visualisation to support day-to-day decisions. We work in short sprints, show early demos, and adjust priorities with the business partner at each review. The main outputs are an end-to-end demonstrator plus a mid-term and final report, with a light replication pack so the approach can be reused on other CDS campuses. The emphasis is on visible progress, straightforward integration, and a user-friendly interface that makes campus energy data easy to understand and act on.

3.1. Work plan

In six months we deliver a practical digital twin for Arnhem's Buiten. We use Agile/Scrum in two-week sprints, releasing an early MVP and iterating. We build breadth-first (end-to-end thin slice), then deepen features based on feedback. Each sprint starts with Planning and ends with a Review/Retrospective; active participation from the BUC Partner is essential including presence at two-week sprint-meetings.

3.1.1. Task list

Task No	Task Title	Lead Participant	Start month	End month
T1	Connecting with SIF	Menzo van Kessel	Februari 2026	May 2026
T2	3D visualization + webapp development	Wouter Damen	Februari 2026	July 2026
T3	Simulation, Prediction & Alerts	Wouter Damen	April 2026	July 2026

3.1.2. Description of individual tasks

Task 1: Connecting with SIF		
Participant	Role:	Person-month
Menzo van Kessel	Lead	2
Wouter Damen	Development & support	0.5
Steven Bronsveld	Validation & organization	0.5
Objectives: - Establish secure, documented connectivity to the HEDGE-IoT SIF/Knowledge Engine (KE), including authentication/authorization and endpoint inventory. - Produce SAREF-aligned campus mappings (v1) for priority assets/telemetry and validate with the Business Use Case partner. - Deliver D1.1 SIF/KE Integration Pack and achieve M1 SIF/KE handshake &		

mapping v1 complete.

Description of work and contribution of individual participants:

- Discover endpoints, set up authn/authz, catalogue feeds; implement a small ingestion/test harness.
- Define configurable SAREF mappings for ≥ 5 asset types; expose a minimal binding API.
- Lead: Menzo; support: Wouter

Task 2: 3D visualization + webapp development

Participant	Role:	Person-month
Wouter Damen	Lead	1.5
Steven Bronsveld	Fullstack developer & Organisation	1.5
Menzo van Kessel	Fullstack developer	3

Objectives:

- Ship role-based dashboards and a 3D digital twin bound to the semantic model with <60 s ingest-to-UI.
- Deliver D2.1 and meet M2 (end-to-end demonstrator).

Description of work and contribution of individual participants:

- Build React/TypeScript app with REST/WebSocket back end; bind 3DBAG/campus geometry to semantic IDs.
- Implement KPI views, layers, and comparison; secure APIs, basic CI/CD, and demo/live modes.
- Lead: Wouter; support: Steven (APIs/validation/organisation), Menzo (binding/mappings).

Task 3: Simulation, Prediction & Alerts

Participant	Role:	Person-month
Wouter Damen	Lead	1
Steven Bronsveld	Fullstack developer & Validation	1
Menzo van Kessel	Fullstack developer	1

Objectives:

- Integrate HEDGE-IoT alerts and provide baseline "what-if" probes (≥ 3 scenarios) with clear assumptions.
- Deliver D3.1 and reach M3 (dashboards + what-if + alerts).

Description of work and contribution of individual participants:

- Standardise alert taxonomy, overlay in 3D/dashboards, add acknowledgement trail.
- Implement lightweight scenario engine and simple forecasts; validate outputs with BUC and document limits.
- Lead: Wouter; support: Menzo (scenario engine), Steven (validation with stakeholders).

3.1.3. List of deliverables

Deliverable No	Deliverable Name	Task No	Nature ^[1]	Dissemination level ^[2]	Delivery date ^[3]
D1.1	SIF/KE Integration (authn/authz, interface, SAREF mapping v1)	T1	Demonstrator	Public	30th of April
D2.1	End-to-end Prototype Demonstrator (live telemetry → 3D with binding API; KPI latency measured)	T2	Demonstrator	Confidential	30th of June
D3.1	Simulation & Alerts (what-if probes + alert visualisation integrated)	T3	Demonstrator	Confidential	30th of July
D0.1	Mid-term report	T1,T2,T3	Report	Confidential	30th of March
D0.2	Final report	T1,T2,T3	Report	Public	31th of July

3.1.4. List of milestones

Milestone No	Milestone Name	Tasks involved	Expected date	Means of verification
M1	SIF/KE handshake & mapping v1 complete	T1	28th of Februari	Successful authn/authz to KE; sample queries return data
M2	Mid-term Review (E2E thin slice)	T1, T2	30th of March	Live telemetry: 3D demo with <60s ingest-to-UI; Decided scope / priorities for T3
M3	Dashboards + what-if + alerts	T2, T3	30th of June	Simulations and visualisation running on live feeds.
M4	Final demonstrator accepted & replication pack	T1, T2, T3	30th of July	Final walkthrough; deployment guide & CDS replication playbook delivered. Ready for second CDS pilot.

3.1.5. Technological Risks

We see a few technological risks:

- **Risk:** Difficulty connecting to the HEDGE-IoT Knowledge Engine and/or Semantic Interoperability Framework.
Explanation: We have not established connection to SIF before. This may prove more complex than expected due to evolving APIs, authentication, or ontology requirements. However, resolving this integration challenge is a core goal of the open call.
Mitigation: We have planned the integration as a non-dependent task to ensure progress in other areas continues regardless of initial connection delays. In addition, our team's academic background and experience with complex ontologies and data models provide a solid basis for resolving potential interoperability issues.
- **Risk:** Incomplete or insufficient data availability from the Arnhem's Buiten campus systems.
Explanation: Some data sources (e.g. metering points, substation interfaces, or historical archives) may not be accessible, complete, or formatted according to the HEDGE-IoT ontology. This could limit the representativeness or functionality of the demonstrator.
Mitigation: We will prioritise integration with the most reliable and complete data streams first and use synthetic or aggregated datasets to fill temporary gaps. The system architecture allows incremental data onboarding, so missing feeds can be added later without redesign.
- **Risk:** Ontology or Schema Mismatch within HEDGE-IoT
Explanation: Because the project is still evolving, we see that changes or inconsistencies between the campus data model and the HEDGE-IoT SIF (e.g. in ETSI SAREF class structures) could lead to integration errors or data loss during semantic mapping.
Mitigation: Mappings will be defined through configurable translation layers rather than hard-coded logic, enabling quick updates if the HEDGE-IoT or SAREF schema evolves.
- **Risk:** Accidental exposure of sensitive infrastructure data
Explanation: During integration or testing, configuration errors or insufficient access controls could unintentionally make campus infrastructure data accessible to unauthorised parties.
Mitigation: We are implementing security measures in line with ISO 27001 principles as part of our ongoing certification process. This includes controlled user access, encrypted data transfer, and secure development environments. Test datasets will be anonymised or aggregated to minimise sensitivity in early stages.
- **Risk:** Limited continuation after project end
Explanation: The project could end directly after the open call if the business case is not considered urgent or financially compelling.
Mitigation: We will initiate continuation discussions early with campus partners to identify opportunities before the project closes.
- **Risk:** Results not deemed valid for operational or business use
Explanation: The demonstrator results may not be considered reliable or accurate enough to guide decisions on grid capacity optimisation or energy-transition strategies.
Mitigation: We will validate outcomes jointly with campus partners, align assumptions with real operational data, and communicate limitations transparently to build confidence in the demonstrator's results. Because we engage in an experimental development, mitigating this risk will be one of the most important aspects of this project. But we are well aware that it takes time to build trust, and during the six months of this open-call, we might not yet achieve this trust for tangible-early-results.

4. Intellectual Property and Ethical Issues

In alignment with the Sub-grant Agreement, Bronscore will retain full IP rights to the developed software. We expect to contribute selected components to open source under permissive licences while protecting proprietary parts through copyright and licensing. Software patents are not applicable. We foresee no ethical issues or certification needs in the short term. There is no active engagement with HEDGE-IoT consortium partners or the relevant business use-case provider.

5. Company description (Third party)

Bronscore B.V. is a startup company located in Nijmegen on the Radboud University Science Park, situated about 30 minutes from the Arnhem's Buiten Electricity Campus. We focus on building technical user-friendly web-applications. We have an academic background in Computing Science, Mathematics and Philosophy. We have previously worked on digital-twins for the manufacturing industry (https://bronscode.nl/hightech_interfaces/). We built a drag-and-drop parametric flow design tool that combines steady-state simulation, DXF drawing generation, a live-catalogue BOM, and automatic compatibility checks/adapters. Furthermore, we are experienced in productising software applications in a SaaS model. (e.g. the Growth Portal, an administrative tool for collaborative subsidies <https://growth-portal.eu/>, this tool might be beneficial for the HedgeIoT consortium as well).

We have been interested in the energy market because of its high societal relevance, complex challenges, and pressing urgency. We would love to contribute to the vision of HEDGE-IoT for creating an interoperable, data-driven, and resilient energy ecosystem. By combining our expertise in web-based digital twins, scalable IoT interfaces, and user-centric visualisation, we aim to help make the framework's results accessible, actionable, and impactful for both industrial partners and local energy communities.

Below, you will find a short introduction of our team:

Steven Bronsveld, MSc BA

After completing his three bachelors (Computing Science, Mathematics and Philosophy), his master's in Mathematical Foundations of Computing Science and publishing in the field of type theory about [Inductive and Coinductive Types](#) and having worked as a freelance software developer, Steven expanded his practice into what is now Bronscore B.V.. As Managing Director (CEO), he combines academic rigor with practical engineering to deliver robust SaaS solutions for clients in education, grant-management, and industry.



Wouter Damen, MSc



Having done a double bachelor's degree in Computing Science and Mathematics, and a master's in Applied Mathematics with Data Science, combining experience in (generative) AI, machine learning theory and functional analysis and having worked as a freelance web designer, he provides Bronscore B.V. with creative product design including both visual and technical solutions. As Operating Director (COO), he also manages the rest of the team and oversees product development from inception to production. Relevant experience also includes a contribution to DNV's [Smart Cable Guard partial discharge classification algorithm](#).

Menzo van Kessel, BSc

Currently finalising his master's in Computing Science at Radboud University, Menzo holds double [bachelor's](#) degrees in Computing Science and Mathematics. He has a strong background in logic, formal verification, and type theory. Having worked with Bronscore for over three years, he has gained extensive experience in TypeScript development and modern web technologies. His combination of theoretical depth and practical expertise makes him a key contributor to Bronscore's focus on building reliable, type-safe SaaS architectures.

